Arithmetic Chiral Homology and Deninger's Motivic Cohomology over $\overline{\mathbb{Z}}$

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Abstract

A Deninger-chiral candidate for motivic cohomology on the Arakelov compactification $\overline{\mathbb{Z}}$ of Spec $\mathbb{Z}$ is defined from a primary arithmetic Heisenberg coefficient datum H^{Ar} and a finite-stage arithmetic-chiral bar complex. Its stable unconditional core is the Tate-Mellin character calculation:

$$\chi_{\mathcal{V}^{\mathrm{prim}}}(s) \;=\; \Gamma_{\mathbb{R}}(s)\prod_p (1-p^{-s})^{-1} \;=\; \Lambda_{\zeta}(s) \qquad (\mathrm{Re}(s)>1),$$

with meromorphic continuation and functional equation supplied by Tate's thesis. The entire Riemann xi-function is

$$\xi_{\mathbb{R}}(s) = \tfrac{1}{2}s(s-1)\Lambda_{\zeta}(s).$$

The Deninger determinant formula, the global zero operator, and the comparison with quantum arithmetic Chern–Simons are recorded as conditional structures.

On this candidate, the Riemann Hypothesis is equivalent to the existence of a Deninger generator Θ_{Ar} whose degree-one spectrum lies on $1/2+i\mathbb{R}$. The equivalent archimedean Tate positivity statement is isolated with the corrected sign convention

$$W_\infty(f*\widetilde{f}) = \frac{1}{2\pi}\int_{-\infty}^{\infty} |\widehat{f}(1/2+it)|^2\left[\mathrm{Re}\,\psi(1/4+it/2)+\log\pi\right]dt \geq 0.$$

Several elementary test-function subclasses satisfy this inequality unconditionally; the full positivity statement remains RH-equivalent.

1 Introduction

1.1 Deninger's cohomological problem

The classical Weil cohomology formalism assigns to a smooth projective variety X over $\mathbb{F}_q$ a graded vector space $H^\bullet(X_{\overline{\mathbb{F}}_q},\mathbb{Q}_\ell)$ with a Frobenius endomorphism F, and recovers the congruence zeta function as

$$Z(X,t) \;=\; \prod_{i=0}^{2\dim X} \det\!\left(1-Ft\,\middle|\,H^i\right)^{(-1)^{i+1}}.$$

Deligne's theorem on the Riemann Hypothesis over finite fields states that the Frobenius eigenvalues on H^i have archimedean absolute value $q^{i/2}$; this is the *purity* of the Weil cohomology, and its consequence on zero placement is the analogue of the classical Riemann Hypothesis for $Z(X,t)$.

Deninger's work, initiated in the 1990s, proposes an analogue of this picture on the arithmetic curve $\overline{\mathbb{Z}}$: a cohomology theory $H^\bullet_\Delta(\overline{\mathbb{Z}})$ with a Frobenius-like endomorphism F whose eigenvalues on H^1_Δ are

the non-trivial zeros of the Riemann zeta function. Under this correspondence, the Riemann Hypothesis itself becomes the purity statement: $|F|$-weight $1/2$ on H^1_Δ forces the zero locus to sit on the critical line.

Two obstacles remain. First, $\mathbb{Z}$ is not a variety over a field of positive characteristic, so there is no literal Frobenius endomorphism to act on a cohomology theory. Second, no cohomology theory has ever been constructed on $\mathbb{Z}$ with the integer-graded finite-dimensional profile that a Weil cohomology demands. Each attempted construction (Deninger's foliated-dynamical systems, Connes' adele class space, Meyer's Schwartz operator) runs into the same wall: a key positivity or spectral hypothesis equivalent to the Riemann Hypothesis.

1.2 The candidate: arithmetic chiral homology

The construction is conditional at the spectral level. It consists of a primary arithmetic Heisenberg coefficient datum H^{Ar}, a finite-stage arithmetic-chiral bar complex, and a conditional Deninger generator Θ_{Ar} whose determinant gives the Deninger Lefschetz formula under the determinant-class hypothesis. The candidate has formal Tate polar lines in degrees 0 and 2 and a degree-one sector whose spectral interpretation is precisely the Hilbert-Polya/Deninger obstruction.

The construction uses operadic data. The coefficient datum is assembled from an archimedean Beilinson–Drinfeld Heisenberg piece and per-prime primary Bost–Connes zero-mode pieces, combined through Tate's pointed restricted tensor product. The bar differential splits as an archimedean chiral differential, a one-dimensional Parshin–Gersten boundary, local primary E_1-bars, and a tame-symbol twisting cochain. On reciprocity-closed finite stages, after fixing the completed convolution Lie algebra and closed spherical units, the square-zero identity is the Maurer–Cartan form of Kato–Parshin reciprocity.

The comparison with Deninger's motivic picture and with arithmetic Chern–Simons is a conditional triangle.

Conjecture 1.1 (conditional triple comparison, informal)**.** *Assume:*

(i) *a Deninger spectral-flow operator Θ_{Ar} on the primary degree-one sector with regularised determinant convention fixed;*

(ii) *a Beilinson–Drinfeld/Costello–Gwilliam comparison between the finite-stage ordered bar and arithmetic chiral homology;*

(iii) *a quantum arithmetic Chern–Simons boundary theory whose boundary trace is the primary class $\mathcal{V}^{\mathrm{prim}}$;*

(iv) *compatible topological completions and dualities.*

Then the following three objects should be canonically equivalent with compatible Θ_{Ar}-actions:

(D) the Deninger motivic target $H^\bullet_\Delta(\mathbb{Z})$;

(O) the operadic Deninger-chiral candidate $H^\bullet_{\mathrm{ch}}(\mathbb{Z}, \mathsf{H}^{\mathrm{Ar}})$;

(C) the boundary chiral homology of the conjectural 3D arithmetic Chern–Simons bulk gauge theory on the étale 3-space $\mathbb{Z}$.

The finite places contribute Bost–Connes Euler factors; the archimedean place contributes the Gaussian Tate factor and the unresolved zero-side operator.

The character-level Koszul shadow of the arithmetic Heisenberg is Tate's functional equation

$$\Lambda_\zeta(s) = \Lambda_\zeta(1-s),$$

rooted in Poisson summation. The Riemann Hypothesis becomes purity of the zero-side generator Θ_{Ar} on degree one. Archimedean Tate positivity is equivalent to this purity once the determinant and spectral-flow hypotheses are imposed; it is proved only on explicit test-function subclasses.

1.3 Mathematical spine

The construction separates three layers. The unconditional layer is the arithmetic coefficient datum, the finite-stage bar complex, and the Tate-Mellin character identity. The conditional layer is the Deninger determinant formula, the global zero generator, the comparison with arithmetic Chern–Simons, and the all-genus extension. The open layer is the determinant-class construction of Θ_{Ar} on the degree-one sector and the full archimedean Tate positivity statement.

2 The Arithmetic Curve

2.1 Arakelov compactification

Let Spec $\mathbb{Z}$ denote the spectrum of the ring of integers. Its closed points correspond bijectively to the rational primes p. We write $\mathcal{Z}$ for the Arakelov datum obtained by adjoining the real archimedean place:

$$\mathcal{Z} \;=\; \operatorname{Spec}\mathbb{Z} \;\sqcup\; \{\infty\}.$$

This is not a scheme-theoretic fibre. It is Spec $\mathbb{Z}$ together with Hermitian data at infinity. The real place carries the archimedean norm $|\cdot|_\infty$ (complexified when analytic chiral fields are used); the finite-prime fibres carry the p-adic norms $|\cdot|_p$; the product formula $\prod_v |x|_v = 1$ for $x \in \mathbb{Q}^\times$ realises $\mathcal{Z}$ as an Arakelov-compact analogue of a smooth projective curve.

2.2 Etale cohomological dimension 3

Mazur (1973, *Notes on étale cohomology of number fields*, Ann. Sci. ÉNS 6) established that Spec $\mathcal{O}_K$ for any number field K has étale cohomological dimension 3 (for the usual torsion coefficient range), not 1 as one might guess from the one-dimensional scheme structure. Artin–Verdier duality supplies a dimension-three duality theory for constructible torsion sheaves, with trace/Yoneda pairings landing in $\mathbb{Q}/\mathbb{Z}$. This is the arithmetic-topological analogue of the orientation pairing of a compact oriented 3-manifold.

This is the starting point of the Mazur–Morishita knots-and-primes dictionary (Mazur 1963 unpublished; Morishita 2002, 2012 arXiv:0904.3399): each finite prime $p \subset \operatorname{Spec}\mathcal{O}_K$ behaves topologically as a knot $K_p \subset \operatorname{Spec}\mathcal{O}_K$, whose tubular neighbourhood is the pro-étale fibre over p. Linking numbers correspond to quadratic residue symbols; knot complements correspond to ramified primes; knot Dehn surgery corresponds to étale covers.

2.3 The arithmetic curve as gluing datum

For the factorisation-algebraic constructions below, $\mathcal{Z}$ enters as a gluing datum: a set of places $|\mathcal{Z}| = \{\infty\} \cup \{\text{primes}\}$ together with

- local coefficient spaces per place (archimedean: $\mathbb{C}_\infty$; finite: perfectoid diamond or Iwasawa algebra);
- pointed E_2-monoidal categories $\mathcal{C}_v$ per place (archimedean: chiral algebra; finite: Iwahori Hecke category);

- pointed spherical units $u_v \in C_v$ (archimedean: unit vacuum of the BD chiral algebra; finite: monoidal unit of the Hecke category);
- boundary maps $d_{v,w}$ reflecting Parshin–Gersten tame-symbol reciprocity between places.

This stratified datum is the *two-dimensional arithmetic surface datum* $\mathfrak{S}_2^{\mathrm{Ar}}$. Its operadic dimension is 2 in the restricted-tensor model; its étale cohomological dimension is 3 in the Mazur–Artin–Verdier sense. The archimedean boundary circle is a heuristic dictionary; no compactified étale homotopy type is asserted.

3 Arithmetic Chiral Algebras

3.1 Beilinson–Drinfeld chiral algebras recalled

A chiral algebra on a smooth complex curve X is a right $\mathcal{D}_X$-module $\mathcal{A}$ equipped with a chiral multiplication

$$\mu\colon j_* j^*(\mathcal{A} \boxtimes \mathcal{A}) \longrightarrow \Delta_! \mathcal{A}$$

where $j\colon (X \times X) \setminus \Delta \hookrightarrow X \times X$ is the open complement of the diagonal $\Delta\colon X \hookrightarrow X \times X$. The chiral multiplication is required to be commutative and to satisfy a Jacobi identity on X^3. Equivalently, a chiral algebra is a factorisation algebra on X: a compatible system of quasi-coherent sheaves $\mathcal{A}_S$ on the Ran space $\mathrm{Ran}(X) = \bigsqcup_n X^n/S_n$ satisfying factorisation on disjoint configurations.

A vertex algebra V gives rise to a chiral algebra on $\mathbb{C}$: the $\mathcal{D}$-module $V \otimes \Omega^1_{\mathbb{C}}$ carries a chiral multiplication determined by the OPE. Beilinson and Drinfeld (2004) proved this correspondence is an equivalence of categories between vertex algebras and chiral algebras on the formal disk.

3.2 Extension to $\mathbb{Z}$

A classical chiral algebra requires a smooth curve over a field. The arithmetic curve $\mathbb{Z}$ fails this hypothesis: the finite fibres $\mathrm{Spec}\,\mathbb{Z}_p$ are not smooth over a base field, and there is no literal chiral OPE at a finite prime. The correct arithmetic replacement is place-by-place, with place-local chiral data assembled globally through a restricted tensor product.

Definition 3.1 (arithmetic chiral coefficient datum on $\mathbb{Z}$)**.** An *arithmetic chiral coefficient datum* on $\mathbb{Z}$ is a datum

$$\mathsf{A}^{\mathrm{Ar}} = (\mathsf{A}_\infty, \{\mathsf{A}_p\}_p, \{u_v\}_v, \tau, \mathcal{T})$$

consisting of:

(i) an archimedean chiral algebra A_∞ in the Beilinson–Drinfeld sense on $\mathbb{C}_\infty$;

(ii) for each prime p, a locally-constant pointed E_1-algebra A_p in a chosen completed linear category over the p-adic chart, equipped with a continuous Euler/Frobenius endomorphism ψ_p;

(iii) for each place v, a pointed spherical unit $u_v \in \mathsf{A}_v$ (the monoidal unit at finite v; the BD vacuum at $v = \infty$);

(iv) a Parshin–Gersten twisting cochain τ with values in the completed convolution Lie algebra of the finite-stage local bars;

(v) a filtered transition system $\mathcal{T}$ whose maps insert closed spherical units and preserve τ.

The formal global object is the pointed restricted tensor product

$$\mathsf{A}^{\mathrm{Ar}} \;=\; \mathsf{A}_\infty \,\widehat{\otimes}\, \widehat{\bigotimes}_p{}' \,\mathsf{A}_p$$

in the sense that all but finitely many tensor factors are the pointed units. Calling this object a genuine chiral algebra is conditional on a comparison theorem with arithmetic factorisation homology.

Remark 3.2 (what is not required). The definition does not require a literal chiral OPE at a finite prime, and does not require a global $\mathcal{D}$-module structure. At the archimedean place the OPE is honest; at a finite prime the local structure is E_1 with Euler/Frobenius enhancement. The asymmetry archimedean/finite is structural and unavoidable.

3.3 The arithmetic Heisenberg

The central example for the Deninger-chiral candidate is the arithmetic Heisenberg coefficient datum.

Construction 3.3 (arithmetic Heisenberg H^{Ar})**.** Let $\mathcal{H}_\infty$ be the standard chiral Heisenberg vertex algebra on $\mathbb{C}_\infty$: the vertex algebra freely generated by a single mode $\alpha_\infty(z)$ with OPE

$$\alpha_\infty(z)\,\alpha_\infty(w) \;\sim\; \frac{1}{(z-w)^2},$$

central charge $c_\infty = 1$.

For each prime p, take the primary unramified Bost–Connes zero-mode factor

$$\Lambda_p \;=\; \bigoplus_{r\geq 0} \mathbb{C}\,\delta_{p^r}, \qquad N_p\delta_{p^r} = r\log(p)\,\delta_{p^r}.$$

Its local character is

$$\mathrm{Tr}_{\Lambda_p}(e^{-sN_p}) = \sum_{r\geq 0} p^{-rs} = (1-p^{-s})^{-1} \qquad (\mathrm{Re}(s) > 0).$$

The endomorphism $\psi_{p,t} = e^{-tN_p}$ is the finite-place Euler flow. It is not identified here with an automorphism of the full local Iwasawa algebra $\mathbb{Z}_p[[\mathrm{Gal}(\mathbb{Q}_p^{\mathrm{ab}}/\mathbb{Q}_p)]]$; ramified and wild local data require a separate comparison.

The arithmetic Heisenberg is

$$\mathsf{H}^{\mathrm{Ar}} \;=\; \mathcal{H}_\infty \,\widehat{\otimes}\, \widehat{\bigotimes}_p{}' \,\Lambda_p$$

with spherical units $u_\infty = |0\rangle_\infty$ (BD vacuum of $\mathcal{H}_\infty$) and $u_p = 1_p$ (unit of Λ_p).

3.4 The primary specialisation

The primary trace class $\mathcal{V}^{\mathrm{prim}}$ is the diagonal Tate-Mellin specialisation of H^{Ar}. Its comparison with the Hochschild trace of the global arithmetic Iwahori Hecke category for $\mathrm{GL}_2/\mathbb{Q}$ is a conjectural Langlands-categorical comparison, not part of the definition used in this standalone.

Definition 3.4 (primary arithmetic trace class)**.** Let $\mathcal{V}^{\mathrm{prim}}$ denote the primary adelic Heisenberg trace candidate obtained by imposing the diagonal character:

$$\chi_{p,s}(\delta_{p^r}) = p^{-rs}, \qquad \chi_{\infty,s}(x) = e^{-\pi x^2}|x|^{s-1}\,dx.$$

The finite character is $\prod_p (1 - p^{-s})^{-1}$ and the archimedean Gaussian Mellin character is $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$.

Remark 3.5. $\mathcal{V}^{\mathrm{prim}}$ has the meromorphic completed zeta $\Lambda_{\zeta}(s)$ as its character before polar correction. It has no vertex-algebra structure and no rational graded dimension. The possible Hochschild-trace interpretation through the global Iwahori Hecke category remains a comparison conjecture.

4 The Arithmetic-Chiral Bar Complex

The candidate chiral homology of H^{Ar} is computed by an arithmetic-chiral bar complex, assembled as a filtered colimit of finite-stage bars in the chosen completed linear category.

4.1 Finite-stage bars

Definition 4.1 (finite-stage arithmetic bar)**.** Let $S = S_f \cup \{\infty\}$ be a finite subset of $|\mathbb{Z}|$ containing the archimedean place. Assume S is closed under the Parshin–Gersten boundaries of the finite set of symbols used in the stage, and assume the transition maps insert closed spherical units. The finite-stage arithmetic bar on the abelian primary scope is

$$B_S^{\mathrm{Ar}}(\mathsf{H}^{\mathrm{Ar}}) \;=\; B^{\mathrm{ch}}(\mathcal{H}_\infty) \,\widehat{\otimes}\, C^{\bullet}_{\mathrm{PG},S}(\mathbb{Z}) \,\widehat{\otimes}\, \widehat{\bigotimes}_{p\in S_f} B_{E_1}^{\Lambda_p,\mathrm{prim}}$$

where:

- $B^{\mathrm{ch}}(\mathcal{H}_\infty)$ is the Beilinson–Drinfeld chiral bar of the archimedean Heisenberg;
- $C^{\bullet}_{\mathrm{PG},S}(\mathbb{Z})$ is the one-dimensional Parshin–Gersten complex on S, with degree 0 rational-symbol part and degree 1 finite-place tame-symbol part;
- $B_{E_1}^{\Lambda_p,\mathrm{prim}}$ is the primary E_1-bar of the local Bost–Connes zero-mode factor.

Transition maps $B_S^{\mathrm{Ar}} \to B_{S'}^{\mathrm{Ar}}$ for $S \subset S'$ insert spherical units at the new places.

4.2 The signed total differential

Fix an order $\infty < p_1 < p_2 < \cdots < p_r$ on S. The signed total differential on B_S^{Ar} is

$$D_S \;=\; d_\infty \;+\; (-1)^{|\cdot|_\infty}\, d_{\mathrm{PG},S} \;+\; \sum_{j=1}^{r} (-1)^{|\cdot|_\infty + |\cdot|_{\mathrm{PG}} + \sum_{i<j} |\cdot|_{p_i}}\, d_{p_j} \tag{1}$$

where each $d_\bullet$ splits further as:

$$d_\infty = d_{\mathrm{int}} + d_{\mathrm{res}} + d_{\mathrm{dR}},$$
$$d_p = d_{\Lambda,\mathrm{int}} + d^{\mathrm{loc}}_{\mathrm{bar},p}.$$

The tame-symbol operator $\iota(\tau_S)$ is included in the Parshin–Gersten/local cross-term of $d_{\mathrm{PG},S}$: τ_S is a degree-one cochain in the completed convolution Lie algebra whose local component at p evaluates a Steinberg symbol by the tame symbol in $k(p)^\times$.

4.3 Square-zero by reciprocity

Proposition 4.2 ($D_S^2 = 0$ on the primary finite stage)**.** *On the abelian primary Heisenberg scope, after fixing the completed convolution Lie algebra for τ_S and closed spherical units for transitions, the differential D_S in Equation* (1) *satisfies $D_S^2 = 0$ on every reciprocity-closed finite stage S.*

Proof sketch. Expanding D_S^2 produces three families of cross-terms:

(i) $d_\infty^2 + d_{\mathrm{PG}}^2 + \sum_p d_p^2$: vanishes term-by-term. The archimedean $d_\infty^2 = 0$ is classical (Beilinson–Drinfeld). The Gersten $d_{\mathrm{PG}}^2 = 0$ follows from the Gersten resolution. The local $d_p^2 = 0$ is the primary E_1-bar square-zero identity.

(ii) Anticommutators of the local differentials from different local pieces: vanish by the Koszul signs in Equation (1).

(iii) Cross-terms involving the twisting cochain τ_S: assemble into the Maurer–Cartan equation

$$d_{\mathrm{PG}}\,\tau_S + \tau_S\,d_{\mathrm{PG}} + \tfrac{1}{2}[\tau_S, \tau_S] \;=\; 0.$$

The Maurer–Cartan equation for the tame-symbol twisting cochain is the chain-level form of Kato–Parshin reciprocity on the arithmetic curve: the boundary of a principal symbol has total tame symbol 1 after multiplying over the places in a reciprocity-closed stage. □

Corollary 4.3 (arithmetic bar well-defined)**.** *The filtered colimit $B^{\mathrm{Ar}}(\mathsf{H}^{\mathrm{Ar}}) = \operatorname{colim}_S B_S^{\mathrm{Ar}}(\mathsf{H}^{\mathrm{Ar}})$ over reciprocity-closed finite stages $S \in \mathrm{Fin}_{\mathrm{rec}}(|\mathbb{Z}|)$ is a well-defined chain complex in the chosen completed category, with differential D induced from $\{D_S\}$ by compatibility of transition maps with the closed spherical units. No exactness of completed filtered colimits is used here.*

5 Chiral Homology Functor

5.1 Definition and functoriality

Definition 5.1 (arithmetic chiral homology)**.** For an arithmetic chiral coefficient datum A^{Ar} on $\mathbb{Z}$, define

$$H^\bullet_{\mathrm{ch}}(\mathbb{Z}, \mathsf{A}^{\mathrm{Ar}}) \;:=\; H^\bullet\big(B^{\mathrm{Ar}}(\mathsf{A}^{\mathrm{Ar}}),\, D\big).$$

$H^\bullet_{\mathrm{ch}}(\mathbb{Z}, -)$ is functorial for strict morphisms of coefficient data: the local maps must preserve pointed units, finite-stage transition maps, Euler/Frobenius endomorphisms, Parshin–Gersten boundary maps, the twisting cochain, and the chosen completions. General functoriality for arbitrary local morphisms is not asserted.

5.2 Cohomological dimension

Conjecture 5.2 (Deninger dimension profile)**.** *On the primary arithmetic Heisenberg candidate, after replacing local bars by the bounded primary Koszul model and proving the associated spectral sequence collapses, one expects*

$$H^i_{\mathrm{ch}}\big(\mathbb{Z}, \mathsf{H}^{\mathrm{Ar}}\big) \;=\; 0 \qquad \textit{for all } i \geq 3.$$

Remark 5.3. The statement is not true for arbitrary arithmetic chiral coefficient data: ordinary E_1 and cyclic bars are generally unbounded. The conjecture is a bounded-primary assertion, not a formal consequence of the definition.

Remark 5.4. This matches Mazur's étale cohomological dimension 3: the operadic construction sees degrees 0, 1, 2; the Artin–Verdier fundamental class sits at degree 3 in the étale cohomology but is not visible operadically. The operadic theory is (after a degree shift) the *degree-reduced* avatar of the étale cohomology.

5.3 Tate lines

Definition 5.5 (formal Tate polar lines)**.** The Deninger-chiral candidate contains formal Tate polar lines

$$\mathbb{C} \cdot |0\rangle, \qquad \mathbb{C} \cdot |\infty\rangle,$$

placed in degrees 0 and 2 and corresponding to the poles of $\Lambda_\zeta(s)$ at $s = 0$ and $s = 1$. Identifying these formal lines with actual cohomology groups is part of the Deninger dimension profile.

5.4 The degree-one bulk

The interesting cohomology sits in degree one:

$$H^1_{\mathrm{ch}}(\mathbb{Z}, \mathsf{H}^{\mathrm{Ar}})$$

is the unresolved sector. Conditional on a Deninger/Hilbert-Polya spectral-flow operator Θ_{Ar} and a determinant-class topology, its spectrum should be the multiset of non-trivial zeros of $\xi_{\mathbb{R}}$. Order-one entireness supplies a Hadamard regularisation of the determinant; it does not make the middle sector finite-dimensional.

6 Tate–Mellin Character and the Determinant Convention

6.1 Primary character identity

The stable character calculation is Tate-Mellin, not a theorem of Zhu theory. The full Heisenberg vertex algebra has the ordinary bosonic Fock character. The zeta character appears only after passing to the primary Gaussian/Bost–Connes sector.

Theorem 6.1 (primary Tate–Mellin character)**.** *For* $\mathrm{Re}(s) > 1$*, the primary arithmetic trace class satisfies*

$$\chi_{\mathcal{V}^{\mathrm{prim}}}(s) \;=\; \Gamma_{\mathbb{R}}(s) \prod_p (1 - p^{-s})^{-1} \;=\; \Lambda_\zeta(s).$$

Tate's thesis supplies the meromorphic continuation and the functional equation $\Lambda_\zeta(s) = \Lambda_\zeta(1-s)$*. The entire function is* $\xi_{\mathbb{R}}(s) = \frac{1}{2}s(s-1)\Lambda_\zeta(s)$*.*

Proof. At a finite prime, Definition 3.4 gives

$$\chi_{p,s} = \mathrm{Tr}_{\Lambda_p}(e^{-sN_p}) = (1 - p^{-s})^{-1}.$$

At the archimedean place, the Gaussian Mellin transform gives

$$\chi_{\infty,s} = \int_{\mathbb{R}} e^{-\pi x^2} |x|^{s-1}\, dx = \pi^{-s/2}\Gamma(s/2) = \Gamma_{\mathbb{R}}(s).$$

The Euler product converges for $\mathrm{Re}(s) > 1$ and equals $\Lambda_\zeta(s)$. Tate's Poisson summation argument gives continuation and the symmetry $s \leftrightarrow 1 - s$. □

Remark 6.2. An arithmetic Zhu correspondence would compare this Mellin character with a q-graded trace on a suitable archimedean vertex-algebraic primary sector. The theorem above uses only the Mellin character. The primary Virasoro central regularisation is a separate residual obligation.

6.2 Lefschetz reading

The determinant-trace identity for the arithmetic chiral homology rewrites the character as a Lefschetz number.

Conjecture 6.3 (conditional Deninger determinant identity). *Assume that the Deninger-chiral candidate carries a closed operator Θ_{Ar} with determinant-class regularisation, with formal polar lines in degrees 0 and 2 and middle spectrum the zero multiset. Then*

$$\Lambda_\zeta(s) \;=\; C\,\frac{\det_\infty\big(s - \Theta_{\mathrm{Ar}} \,\big|\, H^1_{\mathrm{ch}}(\mathbb{Z}, \mathsf{H}^{\mathrm{Ar}})\big)}{\det_\infty\big(s - \Theta_{\mathrm{Ar}} \,\big|\, H^0_{\mathrm{ch}}(\mathbb{Z}, \mathsf{H}^{\mathrm{Ar}})\big)\,\det_\infty\big(s - \Theta_{\mathrm{Ar}} \,\big|\, H^2_{\mathrm{ch}}(\mathbb{Z}, \mathsf{H}^{\mathrm{Ar}})\big)},$$

with H^0 contributing the factor s and H^2 contributing $s - 1$ under the chosen normalisation. Equivalently,

$$\xi_{\mathbb{R}}(s) = \tfrac{1}{2} s(s-1)\Lambda_\zeta(s)$$

is the polar-corrected entire determinant.

Remark 6.4. The determinant convention is additive in $s - \Theta_{\mathrm{Ar}}$, as in Deninger's formulation. The periodic multiplicative determinant convention is not used here.

7 Koszul Duality and the Functional Equation

7.1 Arithmetic Koszul pair

The arithmetic Heisenberg admits a Koszul dual, the arithmetic Heisenberg cobar.

Definition 7.1 (arithmetic Heisenberg Koszul dual). The Koszul dual of H^{Ar} is the cobar construction

$$(\mathsf{H}^{\mathrm{Ar}})^! \;:=\; \Omega^{\mathrm{ch,Ar}}(B^{\mathrm{Ar}}(\mathsf{H}^{\mathrm{Ar}})),$$

with structure maps inherited from the bar by the Beilinson–Drinfeld cobar procedure applied locally at each place and assembled through the Kato–Parshin twisting cochain.

Proposition 7.2 (primary Koszul duality, conditional). *On the abelian primary Heisenberg scope, assuming the completed bar/cobar categories satisfy the Positselski hypotheses, the expected equivalence has the form*

$$D^{\mathrm{b}}\big(\mathsf{H}^{\mathrm{Ar}}\text{-mod}\big) \;\simeq\; D^{\mathrm{b}}\big((\mathsf{H}^{\mathrm{Ar}})^!\text{-comod}\big)^{\mathrm{op}}$$

with shift and duality natural at each place.

7.2 The self-duality identity

Theorem 7.3 (character-level Koszul shadow)**.** *On the primary diagonal, the bar character is $\Lambda_\zeta(s)$ and the cobar character is $\Lambda_\zeta(1-s)$. Therefore the character-level self-duality identity*

$$\kappa_{\mathsf{G}}^{\mathrm{Ar}} \;+\; (\kappa^!)_{\mathsf{G}}^{\mathrm{Ar}} \;=\; 0$$

is the Tate functional equation

$$\Lambda_\zeta(s) \;=\; \Lambda_\zeta(1-s).$$

Proof. The bar character is the Tate-Mellin character of Theorem 6.1. The cobar dual applies Fourier transform on $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$ and sends the diagonal character χ_s to χ_{1-s}. Tate's Poisson summation theorem gives equality of the two characters. This is a character-level identity; it does not by itself construct the Deninger zero operator. □

7.3 Poincaré duality reading

In the Deninger framework, the functional equation $\Lambda_\zeta(s) = \Lambda_\zeta(1-s)$ is read as Poincaré duality on $\mathbb{Z}$: the shift $s \mapsto 1-s$ reflects the degree symmetry $H^i \leftrightarrow H^{2-i}$.

Conjecture 7.4 (Poincaré duality and Koszul self-duality)**.** *Assuming a Poincaré pairing compatible with Tate twist and with the zero operator,*

$$\Theta_{2-i} = 1 - \Theta_i^\vee,$$

the chain-level Poincaré duality of the candidate is conjecturally equivalent to the arithmetic Koszul self-duality $\kappa^{\mathrm{Ar}} + (\kappa^!)^{\mathrm{Ar}} = 0$.

8 Finite-Stage Euler Operators and the Global Zero Generator

8.1 Per-place generators

Definition 8.1 (per-place Euler operators)**.** At each prime p, the primary finite-place operator is the Bost–Connes zero-mode Hamiltonian N_p on Λ_p, with $N_p\delta_{p^r} = r\log(p)\delta_{p^r}$. Its semigroup e^{-tN_p} gives the local Euler factor and is the finite-place input to the candidate.

At the archimedean place, the unconditional operator is the infinitesimal scaling operator entering the Gaussian Mellin transform. Any conjugacy with a Virasoro zero mode $L_0 - c/24$ is conditional on the primary Virasoro comparison.

8.2 Global assembly

Definition 8.2 (finite-stage Euler operator and global hypothesis)**.** For a finite stage $S = S_f \cup \{\infty\}$ define the finite-stage Euler operator

$$N_S \;=\; N_\infty + \sum_{p\in S_f} N_p$$

on the finite tensor product. The global Deninger generator Θ_{Ar} is a conditional operator obtained after choosing a completed domain, determinant regularisation, and spectral-flow comparison to the zero side.

Proposition 8.3 (finite-stage chain-level commutation, conditional). *Assume the local primary bars are N_p-equivariant, the archimedean chiral bar is scaling-equivariant, and the tame-symbol twisting cochain τ_S is equivariant under the finite-stage Euler action. Then*

$$[N_S, D_S] = 0$$

on $B_S^{\mathrm{Ar}}(\mathsf{H}^{\mathrm{Ar}})$.

Proof sketch. Place-local commutation is built into the local primary bars and into the scaling equivariance of the archimedean chiral bar. The remaining term is the cross-term τ_S; the stated equivariance hypothesis is exactly the assertion that the Parshin–Gersten tame-symbol operator has degree zero for the finite-stage Euler action. The signs in Equation (1) then give the displayed commutator. □

Corollary 8.4. *Under the same hypotheses and after passing to a compatible completion, the conditional generator Θ_{Ar} descends to the Deninger-chiral candidate homology.*

9 The Triple Identification Theorem

9.1 The three readings

On the primary Heisenberg scope, three *a priori* distinct languages point to the same desired object. The comparison is conditional.

Conjecture 9.1 (conditional triple identification). *Assume the spectral-flow operator Θ_{Ar}, the ordered-bar/chiral-homology comparison, the profinite quantum arithmetic Chern–Simons boundary theory, and the necessary topological dualities. Then there should be canonical isomorphisms of graded vector spaces, compatible with Θ_{Ar}:*

$$H_\Delta^\bullet(\mathrm{Z}) \;\cong\; H_{\mathrm{ch}}^\bullet(\mathrm{Z}, \mathsf{H}^{\mathrm{Ar}}) \;\cong\; \mathrm{HH}_\bullet^{\mathrm{ch}}(\partial\,\mathrm{ACS}^{\mathrm{qu}}_{k^{\mathrm{Ar}}}).$$

The finite Euler factors are supplied by the Bost–Connes primary pieces; the zero-side spectrum and global determinant are part of the hypothesis.

9.2 Dependency list

The comparison splits into three conditional chain-level identifications.

(D) ≅ (O): Deninger's axioms supply the determinant shape of the missing cohomology. The arithmetic-chiral bar $B^{\mathrm{Ar}}(\mathsf{H}^{\mathrm{Ar}})$ packages these axioms. The comparison becomes theorem-grade only after the spectral-flow operator and determinant-class topology are constructed.

(O) ≅ (C): The boundary chiral homology of a 3D gauge theory, in the ordered-bar presentation, is expected to be a filtered colimit of bar complexes indexed by configurations of closed points. For a quantum arithmetic Chern–Simons theory with boundary trace equal to $\mathcal{V}^{\mathrm{prim}}$, the ordered bar at each configuration should match B_S^{Ar}. This is a boundary comparison hypothesis.

(D) ≅ (C): would follow by composition of the above conditional comparisons.

Deninger target	Chiral candidate	3D ACS reading
$H^\bullet_\Delta(\mathbb{Z})$	$H^\bullet_{\mathrm{ch}}(\mathbb{Z}, \mathsf{H}^{\mathrm{Ar}})$	$\partial\,\mathrm{ACS}^{\mathrm{qu}}$ chiral homology
Zero generator Θ	finite-stage N_S plus Θ_{Ar} hypothesis	Wilson loop around K_p; conditional surface at ∞
Lefschetz determinant	Tate–Mellin character plus determinant hypothesis	Boundary partition function, conditional
Poincaré duality	Koszul self-duality, conditional	3D framed bordism duality, conditional
Purity	$\mathrm{Spec}(\Theta_{\mathrm{Ar}}) \subset 1/2 + i\mathbb{R}$ on H^1	Partition-function spectral theorem, conditional
RH	Archimedean A-TP	Partition-function zeros on critical line
Weil H^1 bulk	H^1_{ch} with spectral-flow hypothesis	Bulk state space on K_p-complement
$q^{i/2}$ weights	p^s-scaling per place	Chern–Simons level k^{Ar}
Tate poles	H^0, H^2 Tate lines	Vacuum and conformal insertion

Table 1: Conditional dictionary between three presentations of the desired motivic cohomology of $\mathbb{Z}$.

9.3 Dictionary table

10 The 3D Arithmetic Chern–Simons Bulk

10.1 Kim's classical arithmetic Chern–Simons

Kim (2015, arXiv:1510.05818) constructs classical arithmetic Chern–Simons: for a finite gauge group A, a cocycle class $c \in H^3(BA, \mathbb{R}/\mathbb{Z})$, and a ring of integers $\mathcal{O}_K$, the classical arithmetic Chern–Simons assigns to each Galois representation $\rho\colon G_K \to A$ the phase

$$\mathrm{ACS}_c(\rho) \;=\; \exp\Big(2\pi i \int_{\mathrm{Spec}\,\mathcal{O}_K} c(\rho)\Big) \;\in\; U(1).$$

The unexponentiated action lies in $\mathbb{R}/\mathbb{Z}$. The integral is defined using the Artin–Verdier duality pairing on $\mathrm{Spec}\,\mathcal{O}_K$ (cohomological dimension 3).

10.2 Quantum extension

The conditional comparison uses a quantum extension of Kim's classical theory: a 3D gauge theory with continuous gauge group G (or its arithmetic avatar) whose path integral assigns a complex amplitude to each Galois representation.

Conjecture 10.1 (quantum arithmetic CS axioms)**.** *Let G be an arithmetic Langlands gauge group. A quantum arithmetic Chern–Simons extension in the form needed here is a conjectural 3D topological field theory on the arithmetic 3-space whose:*

- state space $Z(\Sigma)$ *on a 2-boundary Σ is the arithmetic chiral algebra of the boundary WZW at level k^{Ar};*
- Wilson-loop operator $W(K, \rho)$ *around a link $K \subset \mathbb{Z}$ in representation ρ is the holonomy of the bulk connection;*
- partition function $Z(\mathbb{Z})$ *on the closed arithmetic 3-space is a regularised integral over connections.*

For the primary Heisenberg setup, $Z(\Sigma_\infty) = \mathcal{V}^{\mathrm{prim}}$ at the archimedean boundary circle, and the partition function should recover the polar-corrected determinant $\xi_{\mathbb{R}}$ after the Tate-Mellin character and Deninger determinant hypotheses are imposed.

Remark 10.2. Classical finite-gauge ACS is rigorous (Kim 2015). The quantum extension above is conjectural. It is recorded here as the gauge-theoretic side of the conditional triple comparison, not as an established construction.

10.3 Wilson-loop Frobenius identity

Proposition 10.3 (finite-level unramified Wilson insertion)**.** *At finite level, for an unramified representation ρ_N of the local Galois quotient, the Wilson insertion around the prime knot K_p is the class function*

$$W_N(K_p, \rho_N) \;=\; \operatorname{tr} \rho_N(\mathrm{Frob}_p).$$

This is the finite unramified holonomy statement: Wilson loops read holonomy of the bulk connection, and under the Morishita knots-and-primes dictionary the unramified holonomy around K_p is $\mathrm{Frob}_p \in G_p/I_p$. The profinite, continuous, ramified, and wild statements require the quantum ACS and local epsilon-factor refinements.

10.4 Archimedean Wilson-surface

At the archimedean place the Frobenius is a Wilson-surface.

Conjecture 10.4 (archimedean Wilson-surface dictionary)**.** *Let Σ_∞ denote the formal archimedean boundary cell in the conjectural quantum ACS theory. The archimedean scaling operator should correspond to a Wilson-surface operator*

$$F_\infty \;=\; W(\Sigma_\infty, \sigma_{\mathrm{area}})$$

where σ_{area} is the area-scaling 2-representation of the arithmetic Langlands gauge 2-group. Conjugacy with a Virasoro zero mode is a separate primary-Virasoro comparison.

11 Purity and the Riemann Hypothesis

11.1 Purity defined

Definition 11.1 (Frobenius purity)**.** For the Deninger-chiral candidate, purity on the middle degree means

$$\mathrm{Spec}\big(\Theta_{\mathrm{Ar}} | H^1_{\mathrm{ch}}(\mathbb{Z}, \mathsf{H}^{\mathrm{Ar}})\big) \subset \frac{1}{2} + i\mathbb{R}.$$

Finite-place Bost–Connes unitarity is a separate local statement; it is not a substitute for construction of the global zero operator Θ_{Ar}.

11.2 Finite-prime purity

Proposition 11.2 (finite-prime Bost–Connes unitarity)**.** *For every prime p, the one-parameter unitary group e^{itN_p} on the primary Bost–Connes zero-mode factor has spectrum contained in the unit circle. Its heat trace is the local Euler factor $(1 - p^{-s})^{-1}$.*

Proof. The generator N_p is self-adjoint on the primary Hilbert completion with eigenvalues $r \log p$ for $r \geq 0$. Hence e^{itN_p} is unitary. The heat trace is $\sum_{r\geq 0} e^{-sr\log p} = (1-p^{-s})^{-1}$. □

11.3 Archimedean purity

Archimedean purity is the outstanding residual.

Theorem 11.3 (RH ⇔ zero-generator purity, conditional)**.** *Assume the Deninger determinant identity of Conjecture 6.3: the middle operator Θ_{Ar} is determinant-class and its spectrum is the non-trivial zero multiset of $\xi_{\mathbb{R}}$. Then the Riemann Hypothesis is equivalent to purity*

$$\mathrm{Spec}(\Theta_{\mathrm{Ar}}|H^1) \subset \frac{1}{2} + i\mathbb{R}.$$

Proof. Under the hypothesis, each eigenvalue of Θ_{Ar} is a non-trivial zero ρ of $\xi_{\mathbb{R}}$, with multiplicity. The displayed purity condition is exactly $\mathrm{Re}(\rho) = 1/2$ for every non-trivial zero. □

11.4 Archimedean purity as positivity

Archimedean purity can be reformulated as an explicit positivity statement.

Theorem 11.4 (archimedean Tate positivity)**.** *Archimedean Frobenius purity is equivalent to the* archimedean Tate positivity *(A-TP) statement:*

$$W_\infty(f * \widetilde{f}) \;\geq\; 0 \qquad \textit{for every admissible } f \in \mathrm{PW}(\mathbb{R})$$

where W_∞ is the archimedean contribution to the Weil explicit formula,

$$W_\infty(f * \widetilde{f}) \;=\; \frac{1}{2\pi}\int_{-\infty}^{\infty} |\widehat{f}(1/2+it)|^2 \left[\mathrm{Re}\,\psi(1/4+it/2) + \log\pi\right] dt.$$

Proof sketch. On the scoped admissible test class, the explicit Weil distribution decomposes into finite-prime, archimedean, and polar/trivial-zero pieces. The finite-prime and polar pieces are closed by the corresponding trace-formula and boundary-term inputs on that scoped class. The remaining inequality is exactly the displayed archimedean integral. Full Weil positivity is RH-equivalent; the theorem proves an equivalence of formulations. □

12 A-TP and the Elementary Sub-Classes

A-TP is Riemann-Hypothesis-equivalent. The unconditional result is A-TP on explicit test-function sub-classes, each proved by a distinct method. No linear inclusion order among the classes is claimed.

12.1 The digamma kernel

Define the digamma kernel

$$g(t) \;:=\; \mathrm{Re}\,\psi(1/4+it/2) + \log\pi.$$

Proposition 12.1 (monotonicity + sign-change)**.** *g is even, strictly increasing in $|t|$, with unique positive zero $t_* = 0.8506623044\ldots$.*

Proposition 12.2 (exact identity). *$g(t) = 2\theta'_{\mathrm{RS}}(t) + 2\log\pi$, where $\theta_{\mathrm{RS}}(t) = \mathrm{Im}\,\log\Gamma(1/4 + it/2) - (t/2)\log\pi$ uses the continuous branch with $\theta_{\mathrm{RS}}(0) = 0$.*

12.2 Unconditional sub-classes

Proposition 12.3 (audited A-TP sub-classes). *Each of the following sub-classes of $\mathrm{PW}(\mathbb{R})$ admits an unconditional proof of A-TP on its elements:*

- ***HFD*** *high-frequency dominance: $\int_{|t|>t_*} |\widehat{f}|^2\, g \geq \int_{|t|<t_*} |\widehat{f}|^2 |g|$ (digamma Hurwitz instrument);*
- ***SHF*** *Stirling-four high-frequency, $T_{\mathrm{St4}} \approx 0.873$ enlargement;*
- ***HD*** *Hankel dominance via Binet–Laplace representation;*
- C^+_{Gauss} *Gaussian positive cone at $\lambda \geq \lambda_c \approx 2.52$ (explicit* erfc *series);*
- ***VK instrument*** *test functions whose support and growth allow the Vinogradov–Korobov zero-free strip of width $\eta(T) \asymp (\log T)^{-2/3}(\log\log T)^{1/3}$ to dominate the residual term;*
- $\mathrm{PW}^{\mathrm{TM}}$ *Tate–Mellin sub-class admitting factorisation $f_M(s) = \xi_{\mathbb{R}}(s) \cdot h(s)$ with Hermite symmetry;*
- $\mathrm{PW}^{\mathrm{CP}}$ *Burnol co-Poisson sub-class: Fourier-dual to HFD;*
- ***IE*** *integral-energy sub-class: Slepian–Pollak prolate-spheroidal instrument; unconditionally covers test functions of exponential type $\leq 1/2$.*

The union is a genuine enlargement of the individually proved classes. No quantified coverage theorem for the complement is claimed.

12.3 The structural floor

Proposition 12.4 (structural floor diagnosis). *The elementary reductions of A-TP (digamma Hurwitz, Stirling, Binet, Mellin–Barnes, Hadamard pair-pole, Gram matrix, Guinand–Weil, Nyman–Beurling Möbius, Bost–Connes scattering) leave the same kind of residual: a compact archimedean positivity problem not removed by the elementary transform. This is a diagnosis of these routes, tied to the non-Hausdorff nature of the adele class space $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times}$.*

12.4 Fourier-dual obstructions

Conjecture 12.5 (Fourier-dual bounded-interval obstruction). *The residual of A-TP decomposes into two Fourier-dual bounded-interval obstructions:*

- *spectral-side residual on $t \in [-t_*, t_*]$ with $t_* \approx 0.8507$;*
- *time-side residual on $y \in (0, 1/\pi)$ with $1/\pi \approx 0.318$.*

The two are conjecturally Fourier-dual under the Burnol co-Poisson transform. A theorem requires the residual functional, the transform domain, and the endpoint normalisation. The full obstruction is RH-equivalent.

13 All-Genus Extension

13.1 Arithmetic genus-g data

Definition 13.1 (arithmetic genus-g datum). For $g \geq 0$, an arithmetic genus-g datum is a triple

$$\mathcal{D}_g = (B_g, E_g, \varphi_g)$$

where B_g is the chosen arithmetic base/window, E_g is the cohomological or chiral coefficient object, and φ_g is the finite-place and archimedean Frobenius/scaling datum. Its motivic center and weight are denoted $w_{\mathcal{D}_g}$. A rank-$2g$ Heisenberg model is one realisation of E_g, not the definition of the datum.

13.2 All-genus character identity

Conjecture 13.2 (conditional all-genus Lefschetz character)**.** *Assume the genus-g Deninger package, rank-g Koszul-Petersson duality, finite-place temperedness, arithmetic Chern–Simons comparison, and spectral-flow determinant. Then*

$$\chi\big(\mathcal{V}_g^{\mathrm{prim}}\big)(s) \;=\; \prod_i \det{}_\infty\big(s-\Theta_g \,\big|\, H^i_{\mathrm{ch}}(B_g,E_g)\big)^{(-1)^{i+1}} \;=\; \Lambda_{\mathcal{D}_g}(s),$$

where $\Lambda_{\mathcal{D}_g}$ is the completed motivic L-function of the genus-g datum. For $g=0$, this recovers Theorem 6.3 and the Riemann framework.

13.3 Motivic-weight RH

Conjecture 13.3 (motivic-weight Riemann Hypothesis reformulation)**.** *Under the same hypotheses, the motivic Riemann Hypothesis for $\mathcal{D}_g$ is equivalent to placement of the degree-one zero-generator spectrum on the critical line determined by $w_{\mathcal{D}_g}$.*

14 Residual Obstructions

After the constructions of Sections 3–13, the obligations separate as follows.

Status	Obligation
closed in this paper	primary finite Euler factor and Gaussian Tate factor
closed on finite stages	$D_S^2=0$ on the abelian primary reciprocity-closed bar
conditional	global determinant-class operator Θ_{Ar}
conditional	ordered-bar/chiral-homology comparison
conditional	quantum arithmetic Chern–Simons boundary theory
conditional	archimedean Wilson-surface dictionary and primary Virasoro comparison
open/RH-equivalent	Hilbert–Polya spectral flow and full A-TP
conditional all-genus	rank-g Koszul-Petersson and motivic weight compatibility

14.1 Path to RH

Under the spectral-flow and determinant hypotheses, RH reduces to archimedean Tate positivity (Theorem 11.4). A-TP is a statement about one explicit integral, the archimedean contribution $W_\infty(f*\widetilde{f})$ against the digamma kernel $g(t)=\mathrm{Re}\,\psi(1/4+it/2)+\log\pi$. It is closed unconditionally on the audited test-function sub-classes listed above. The complement remains RH-equivalent.

The arithmetic chiral homology framework isolates A-TP as the residual problem:

- A precise, chain-level-explicit candidate for Deninger's cohomological profile.
- A conditional triple comparison in which the motivic, operadic, and gauge-theoretic languages are conjecturally presentations of the same object.

- An honest isolation of the residual obstruction as a single positivity statement with Fourier-dual symmetric form.
- An unconditional lower bound on the proved sub-class: the listed instruments supply unconditional A-TP on their union.

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